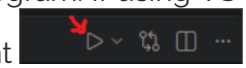
Lab Procedure

Play

Introduction/Setup

1. Launch Quanser Interactive Labs and load the QArm Workspace.
2. Open sp0_play/play_qarm/digital_twin folder. This document will assume VS Code is used.

Position Mode

1. Open **BasicIO_position_mode.py** in VS Code or a text editor of your choice.
2. Run the Program. If using VS code open the python file and press the “run” button at the top right 
    a. The Base light should turn green
    b. The Virtual QArm should move to the home position
    c. Two GUI windows as shown below will appear:”

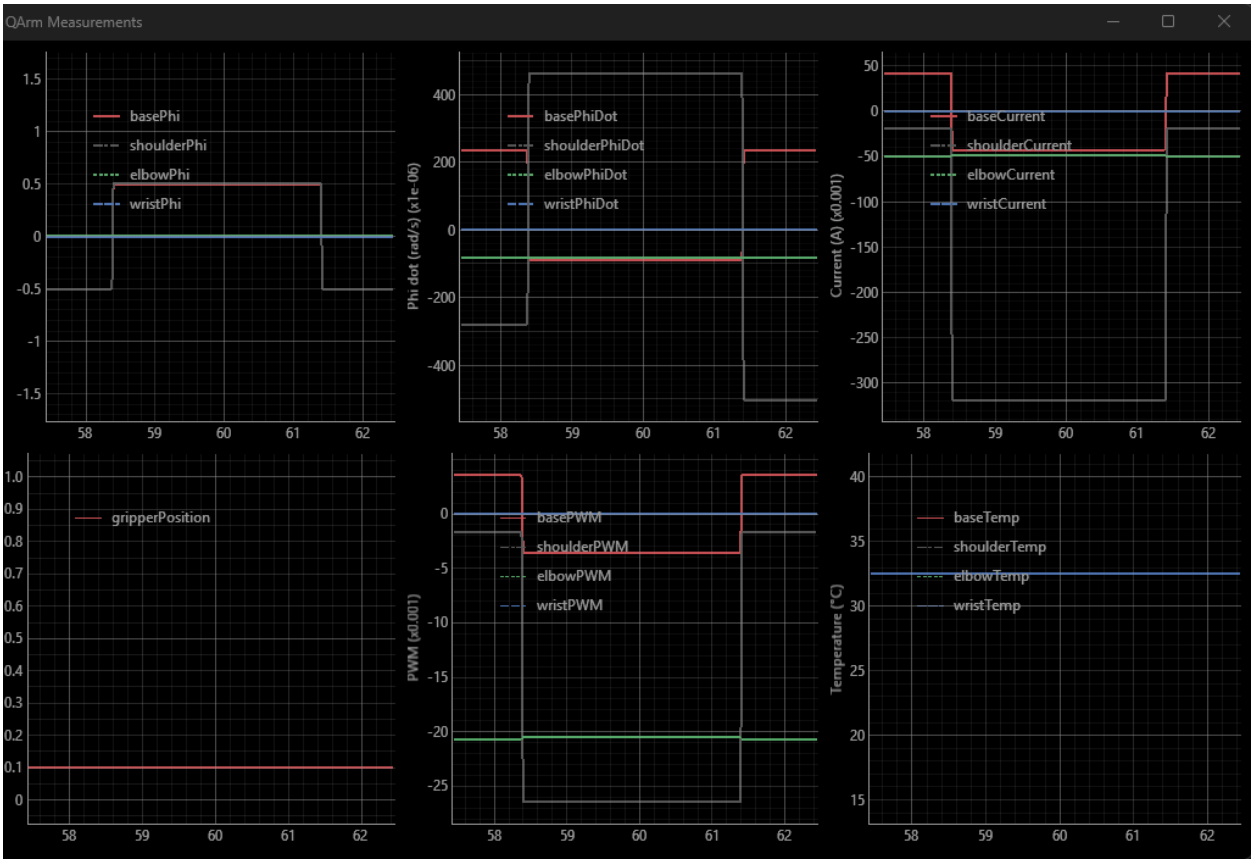


Figure 1: “QArm Measurements” window, which displays info about the arm

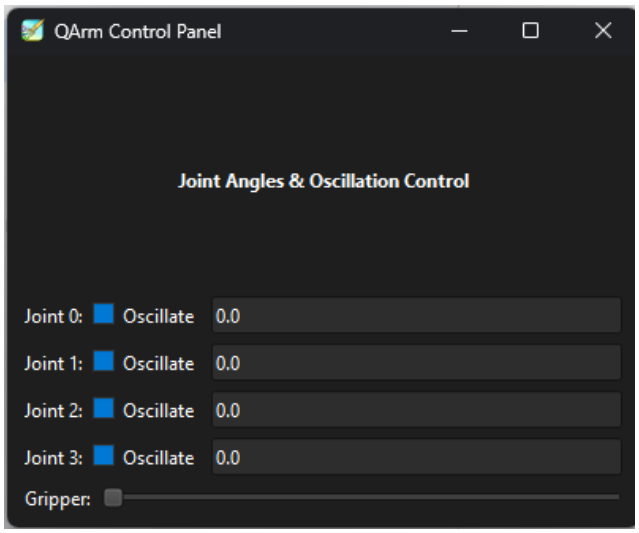


Figure 2: "QArm Control Panel" window, which will allow you to adjust values

3. Using the "QArm Control Panel" window, (Figure 2) adjust the joint positions by replacing the zeros with a number of your choice. This represents the joint angle in radians.

    a. Command small changes (i.e. ±0.25 rad) to determine the positive/negative angle convention

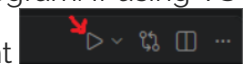
4. Using the gripper slider, adjust the position of the gripper. Under the hood, this is a value that ranges from 0 (open) to 1 (closed)

5. Toggle the oscillate checkbox to send a square wave signal to the desired joint. This will cause the joint to oscillate between +0.5 rad to −0.5 rad with a period of 6 seconds (3 seconds between each movement)

6. For all actions, use the virtual QArm visual and the "QArm measurements" window (Figure 1) to verify behaviour.

7. If no further experiment is required, close both GUI windows to end the program. It is not necessary to close the QLabs window.

PWM Mode

1. Open **BasicIO_pwm_mode.py** in VS Code or a text editor of your choice.
2. Run the Program. If using VS code open the python file and press the "run" button at the top right 
3. A similar UI as position mode will appear. The QArm will start in the home position and begin to slowly top forward.
    a. The "QArm measurements" window is the same and shows the same data.
    b. The "QArm Control Panel" window now controls PWM. Select the –1, 0 or +1 buttons to autofill a PWM command. This corresponds to –12V, 0V and +12V respectively. You can also enter a custom value into the textbox on the right.

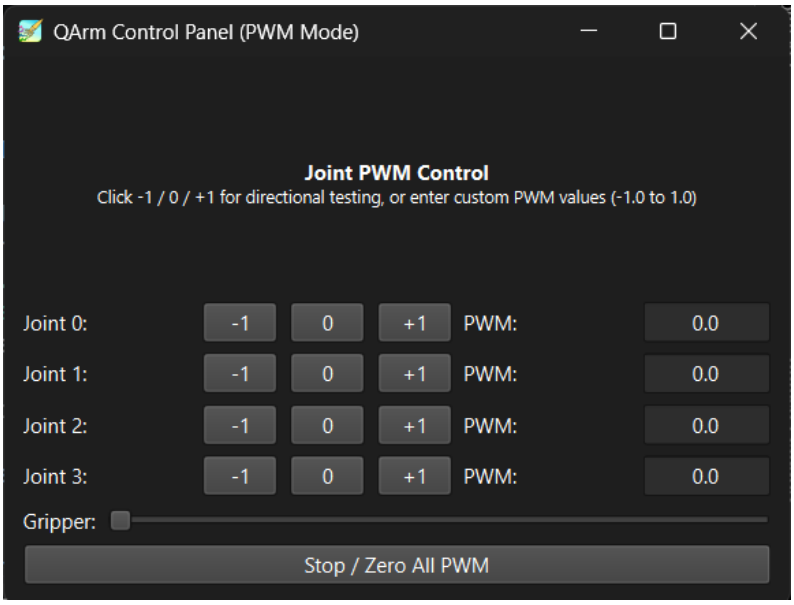


Figure 3: "QArm Control Panel" window for PWM mode, which will allow you to adjust values

4. Toggle commands and verify the joint directions that correspond with positive/negative inputs. Is it the same convention as position mode?
5. Unlike position mode, the PWM bypasses the manipulator's built in controller, and now runs in an open loop fashion. This makes it difficult to navigate and maintain desired positions. To explore this further, try to maintain the home position using manual commands. Note down the PWM values that holds the home position.
6. No further experiment required. Close both windows and Quanser Interactive Labs.